

Multimodality as Bridge or Hurdle in Human–AI Communication

Selective Interpretability in Game-Based Multimodal Tasks

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Abstract

Multimodal communication often conveys meaning beyond literal description, through context, association, and shared human knowledge. This work studies when such multimodal cues act as bridges that support interpretation, and when they become hurdles that make interpretation harder. We introduce a reusable game-based framework that models multimodal communication as a sender–receiver–validator process, with communicative difficulty controlled through alphabets, signatures, and task constraints. We instantiate this framework in an adversarial human–AI captioning task over paired images, where captions are written either to support human interpretation or to remain human-interpretable while becoming harder for AI. Our results show that the AI-aware condition reduces performance for both humans and AI, but also increases divergence between human and AI interpretations. Qualitative observations suggest that humans shift from literal descriptions towards mood, metaphor, and association when writing against AI, revealing expectations about AI limitations. The study provides an exploratory framework for analyzing hidden meaning as a bridge for some receivers and a hurdle for others.

Keywords

multimodal communication, human-AI interaction, interpretability, captioning

1 Introduction

Human communication is inherently multimodal; people routinely combine text, images, symbols, and context in order to convey meanings that go beyond literal description. Despite recent progress in multimodal machine learning, artificial intelligence systems still struggle when meaning depends on abstraction, context, shared background knowledge, or pragmatic interpretation.

For example, consider an image depicting children playing together (Figure 1). A literal description might focus on observable elements such as “two children in a park”. However, the image may also communicate a more abstract concept such as *friendship*. Humans can often infer such non-literal, context-dependent meanings with little effort. In contrast, generative text-to-image systems often reveal limitations when asked to reconstruct abstract concepts such as *friendship*, frequently resorting to stereotypical or overly literal visualizations (Figure 2). This gap highlights a persistent challenge in multimodal machine learning: distinguishing surface-level description from deeper semantic or pragmatic meaning.

The idea of exploiting differences between human and machine interpretation is familiar from CAPTCHAs: tasks are designed around abilities that are assumed to be easier for humans than



Figure 1: A literal scene of playing children, that may also communicate a more abstract meaning such as friendship.



Figure 2: Attempt to generate image using DALL-E 1 [7] using the prompt: *friendship*; illustrating the tendency toward literal or stereotypical visualizations.

for machines. Classic CAPTCHAs often rely on perceptual distortion, visual noise, or recognition tasks that were historically difficult for automated systems. Their underlying logic is to identify an aspect of perception or reasoning that is treated as comparatively human-accessible. In this work, we are interested in a related but more semantic form of this asymmetry: whether hidden or indirect meaning in multimodal communication can serve as such a human-accessible signal.

Captioning provides a natural setting for this question. Some images do not provide one obvious clue that uniquely determines what should be said about them. Instead, they invite multiple possible descriptions: one may describe the visible scene, the mood it evokes, a cultural association, or an anecdotal interpretation.

We study this problem through the notion of *bridges* and *hurdles*. A multimodal cue acts as a bridge when it helps a receiver recover the intended meaning, and as a hurdle when it requires additional inference or contextual knowledge that the receiver may not have. To investigate such effects systematically, we introduce a game-based framework for multimodal communication tasks. Games are useful here because they separate sender, receiver, and verification, while allowing communicative difficulty to be controlled through task mechanics.

Building on this framework, we introduce an adversarial human-AI captioning task over paired images. Human participants produce captions that should remain understandable for other humans while becoming harder for AI to interpret. This allows us to study not only whether hidden meaning can create human-AI divergence, but also what human captioning strategies reveal about expectations of AI limitations.

The contributions of this paper are threefold. First, we frame multimodal communication in terms of bridge-hurdle effects, where the same multimodal signal may support one receiver while obstructing another. Second, we introduce a reusable game-based framework that captures multimodal communication tasks through a shared sender-receiver-validator structure and allows communicative difficulty to be controlled through task mechanics. Third, we instantiate this framework in an adversarial captioning task to study whether hidden meaning can be used as a human-accessible signal, and what human captioning strategies reveal about expectations of AI limitations.

2 Background: Multimodal Communication & Semiotics

We approach multimodality as a problem of communication. The question is not only whether different modalities can be combined, but how a message is formed, constrained, interpreted, and verified by different receivers. This connects directly to the contribution of this work: we use a communication framework to study when multimodal cues help interpretation and when they become difficult to recover. The bridge aspect is familiar from everyday human communication, where images, gestures, or symbols naturally support expression. The hurdle aspect becomes especially visible when communication is intentionally constrained, as in games: the same multimodal richness that supports expression can also introduce ambiguity, hidden associations, or contextual requirements that make interpretation harder. Unlike everyday communication, games allow such hurdles to be deliberately introduced and controlled.

2.1 Multimodality as Bridge or Hurdle

Multimodality refers to the use of multiple representational systems, such as text, images, symbols, audio signals, or spatial arrangements, to convey meaning. Human communication is inherently multimodal: gestures accompany speech, images complement written

language, and symbolic systems allow speakers to communicate concepts that are difficult to express through literal language alone.

From a computational perspective, multimodality is not merely an additional technical capability, but a structural property of communication. It can support communication, but it can also complicate it. In this work, we describe these two roles as *bridge* and *hurdle*.

A modality functions as a *bridge* when it compensates for the limitations of another modality. When a concept is difficult to verbalize directly, an image, drawing, or restricted symbolic clue may make the intended meaning easier to express. In this sense, multimodality can help externalize meanings that would otherwise remain vague, underspecified, or hard to communicate.

At the same time, a modality can become a *hurdle*. Every additional modality introduces another layer of interpretation, and successful understanding depends on whether sender and receiver align on the same associations, abstractions, and contextual cues. Misalignment between modalities or between the communicative abilities of sender and receiver can introduce ambiguity, noise, and interpretation errors. Multimodality therefore does not automatically make communication easier or harder; it changes the conditions under which communication succeeds or fails.

This dual role is especially relevant for human-AI interaction. A hidden or indirect meaning may act as a bridge for one kind of receiver while becoming a hurdle for another. For humans, pragmatic context, cultural background, or shared experience may support interpretation. For AI systems, the same communicative move may remain difficult if it depends on associations that are weakly represented in training data or cannot be recovered from surface-level alignment alone. Multimodality therefore provides a useful setting for investigating not only whether a message can be understood, but also *by whom* it can be understood.

2.2 Communication as Encoding-Decoding in Games

The CAPTCHA idea introduced earlier already has a game-like structure: a challenge is presented, a participant responds, and success is verified. Many other multimodal communication scenarios follow a similar pattern, including *4 Pics 1 Word*, *Movie Emoji Trivia*, the *ESP Game*, *Guess Who?*, and *Codenames*. In such settings, one participant encodes a concept under certain constraints, another participant attempts to interpret it, and a verification step determines success. We therefore treat these scenarios as games in a broad sense: structured communication tasks with roles, constraints, and measurable outcomes.

These games make visible several core questions of multimodal communication: How are encoding and interpretation represented? Which constraints are imposed on the sender and the receiver? How is communication intentionally made easier or harder through task design? What role does multimodality play — as bridge or as hurdle? Which forms of interpretation should remain stable across individuals, cultures, or representational styles?

These considerations motivate a general view of communication as an encoding-decoding process. A sender transforms a target concept into a message, the message passes through a potentially noisy channel, and a receiver interprets it. In multimodal settings,

this noise may arise from ambiguity, cultural differences, limited expressive resolution, or imperfect cross-modal alignment. Importantly, such noise is closely related to communicative difficulty. What makes a task hard is often precisely what makes interpretation uncertain: multiple possible associations, incomplete cues, or a mismatch between what the sender assumes and what the receiver can recover.

Games are especially useful in this context because they make this difficulty controllable. Through task design, one can regulate how much uncertainty is introduced into the communication process: by restricting the available symbols, forcing communication through a particular modality, limiting the form of the answer, or adding mechanics such as time pressure or reduced context. Difficulty is therefore not merely a by-product of the task, but a design parameter that shapes the communicative behavior of sender and receiver.

This point is central to our notion of bridge and hurdle. By controlling communicative difficulty, one can influence whether multimodality primarily supports interpretation or obstructs it. If an additional modality provides cues that compensate for weaknesses in another modality, it acts as a bridge. If the same modality introduces ambiguity, requires hidden associations, or depends on knowledge unavailable to the receiver, it acts as a hurdle. The same multimodal resource can therefore play different roles depending on how the task is designed and how much difficulty is introduced.

In *Codenames: Pictures*, for example, multimodality can act as a bridge because images reduce dependence on language-specific vocabulary and allow associations based on visual structure as well as meaning. At the same time, it can act as a hurdle because pictures are more open to interpretation than words: different players may focus on different details and derive different associations from the same clue. The same multimodal design thus supports some forms of understanding while obstructing others.

2.3 Semiotic Perspective

A central distinction from semiotics helps to clarify these phenomena. **Syntax** concerns the structural form of a message, such as the order of words or the composition of symbols. **Semantics** concerns the literal meaning associated with these forms. **Pragmatics**, in contrast, concerns interpretation in context: shared knowledge, communicative intent, cultural conventions, and the assumptions speakers make about their audience [1, 3].

This distinction is useful for our purposes because the challenge in multimodal communication is not only whether a system can align words and images, but whether it can recover the intended meaning when that meaning depends on hidden association, contextual knowledge, or indirect reference [4].

A related distinction concerns *invariance* and *variance*. In many communication settings, successful encoding should lead to invariant interpretation: the same message should be understood similarly across individuals, cultures, or presentation styles. In other settings, however, variance is not only acceptable but desirable. A sender may intentionally produce a message that remains understandable only to a specific audience or under a specific interpretive frame.

Such selective accessibility becomes especially relevant in competitive or adversarial settings, where a message may function as a bridge for one receiver while becoming a hurdle for another.

We draw on semiotics and related literature as conceptual background for thinking about signs, meaning, and interpretation. However, most of this literature does not directly provide a framework for experimentally varying bridge–hurdle effects in multimodal human–AI communication. The specific combination developed here — modeling multimodal games through controllable difficulty, alphabets, signatures, and replaceable human or AI agents — is therefore treated as a new methodological contribution rather than a direct application of an existing semiotic model.

3 Framework: Multimodal Games as Encoder–Decoder–Validator Systems

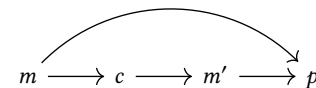
To study the multimodal communication scenarios discussed in the background within a common system, we introduce a modular framework that captures a wide range of games using the same underlying structure. Rather than treating each game separately, the framework abstracts them into a common flow with a sender, a receiver, and a verification step. In this sense, it operationalizes the game-like view of communication introduced earlier: a concept is encoded under constraints, interpreted by another agent, and then evaluated.

The framework uses the following core components that also structure the implementation: *Task*, *Example*, *Alphabet*, *Signature*, *Adapter*, *Agent*, and *Game*; which will be introduced in the upcoming section. These components allow different multimodal games to be instantiated by changing the data, the available symbol sets, the allowed forms of interaction, and the participating agents, while keeping the overall execution logic fixed.

3.1 General Flow Through the Framework

The framework is designed to capture many of the multimodal games discussed in the background within one common structure. Although these games differ in modality, symbol set, and task mechanic, they share the same basic pattern: a sender is given a target message, produces a constrained representation of it, a receiver interprets that representation, and a verification step evaluates the outcome.

Figure 3 illustrates this general flow. At the conceptual level, a single round follows



where m is the original target message, c is the encoded message produced by the sender, m' is the receiver's interpretation, and p is the result returned by the validator.

The framework keeps this overall flow fixed, while allowing the concrete form of communication to vary from task to task. A game may use words, emojis, images, or other multimodal symbols; it may require free-form output, single choices, or restricted combinations; and it may be carried out by human participants, AI models, or a combination of both. What makes these different scenarios

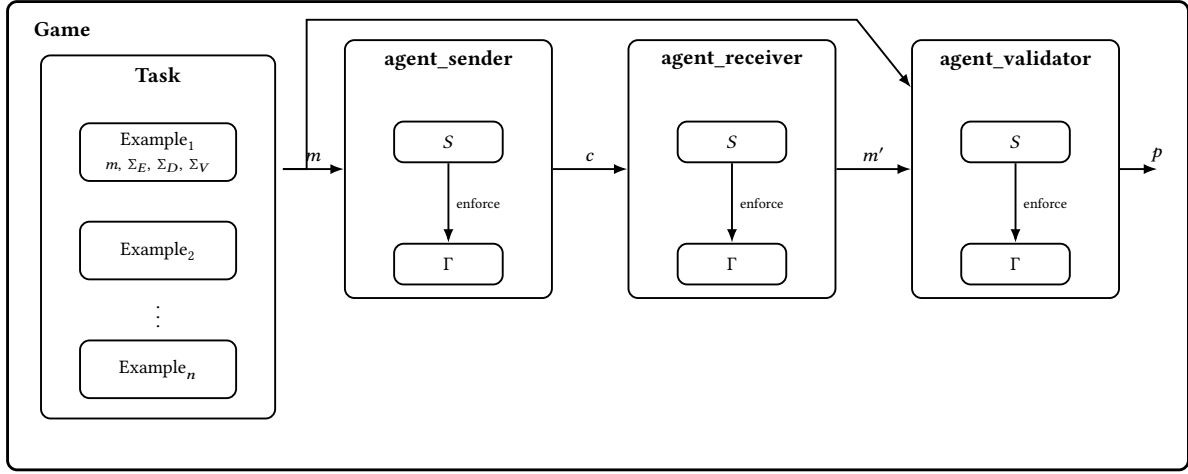


Figure 3: Framework modules and message flow. Each agent combines a signature S and an adapter Γ

comparable is that they can all be represented within the same sender–receiver–validator loop.

This common flow also provides the point at which communicative difficulty can be controlled. The difficulty of a task does not arise only from the target message itself, but from the alphabets and structural constraints through which that message must be expressed and interpreted. The following subsections describe how this is implemented through tasks and examples, through agents with signatures and adapters, and through a shared execution logic.

3.2 Tasks and Examples

A *Task* T is a set of examples that instantiate one particular game mechanic. Each example specifies a target message or concept together with the alphabets that are available in the encoding, decoding, and validation phases. Formally, we write

$$x = (m, \Sigma_E, \Sigma_D, \Sigma_V),$$

where $m \in \mathcal{M}$ is the target message, Σ_E is the encoder alphabet, Σ_D is the decoder alphabet, and Σ_V is the validator alphabet.

This representation makes the framework general enough to capture many different multimodal games using the same overall structure. In one task, Σ_E may consist of emojis and Σ_D of movie titles; in another, Σ_E may be a set of images and Σ_D a set of concepts. The framework therefore does not depend on a particular modality, but on how the task specifies the available forms of communication.

3.3 Agents, Signatures, and Adapters

The active components of the framework are the sender, the receiver, and the validator. Each of them is modeled as an *agent*. An agent contains a *Signature* S and an *Adapter* Γ , but is not reducible to these two components alone: it also implements the control logic that connects the current game phase to the adapter call and the signature check. In other words, the agent determines how the current state is presented, how an output is obtained, and whether that output is valid before the game advances.

The signature specifies which outputs are admissible over a given alphabet. In this sense, it defines the structural rules of the game

mechanic. Typical signatures include:

- single choice: $x \in \Sigma$,
- multiple choice: $X \subseteq \Sigma$,
- word sequence: Σ^+ ,
- fixed-length sequence: Σ^n ,
- permutation: $\text{Perm}(\Sigma)$.

The adapter connects this abstract role to a concrete backend. A human adapter may present text or images in a notebook or web interface and collect manual input. An AI adapter may convert the current state into a prompt, call an external multimodal model, and parse the returned answer. The agent does not contain the AI model itself; the model is treated as an external backend accessed through the adapter. Because the adapter is separated from the signature, the same game structure can be carried out with either human or AI participants.

3.4 Execution Logic

At the conceptual level, the framework follows a fixed three-step flow, which is implemented through the three game phases:

- ENCODING: $\text{agent_sender}(m, \Sigma_E) \rightarrow c$,
- DECODING: $\text{agent_receiver}(c, \Sigma_D) \rightarrow m'$,
- VALIDATING: $\text{agent_validator}(m, m', \Sigma_V) \rightarrow p$.

First, the sender encodes the target message m into a constrained representation c . Second, the receiver interprets this representation and produces an output m' . Third, the validator compares the interpretation with the original target and returns a result p , for example in the form of correctness or a score.

This execution logic remains constant across tasks. What changes from one game to another are the messages, alphabets, the signatures, and the participating agents. As a result, the same framework can capture free-text captioning, emoji-based guessing games, image selection tasks, and other structured multimodal interactions without changing the overall control flow.

3.5 Difficulty Through Alphabets and Signatures

A central property of the framework is that communicative difficulty can be introduced and controlled explicitly. As argued in the background section, games are useful because they allow noise and uncertainty in communication to be turned into a design parameter. In the framework, this happens mainly through the alphabets and signatures.

The alphabet determines *what* can be expressed. The signature determines *how* it may be expressed. Together, they define the communicative search space available to sender and receiver. Restricting the alphabet may force symbolic abstraction; restricting the signature may reduce expressive freedom or make interpretation more ambiguous. In this sense, alphabets and signatures provide the main mechanism through which multimodality can be steered toward acting as a bridge or as a hurdle.

For example, an encoder restricted to a small emoji alphabet must communicate through metaphor and symbolic association rather than literal description. A decoder restricted to a single choice over a finite candidate set may simplify evaluation, but also changes the kind of reasoning required. More generally, changing the alphabets or signatures changes the type and amount of communicative difficulty in the task. It is this controlled difficulty that allows the framework to capture a broad variety of multimodal games while also providing a systematic way to study bridge–hurdle effects.

3.6 Concrete Instantiations

The same framework can instantiate different multimodal games by varying examples, alphabets, signatures, and agents.

4 Images 1 Word. In this setting, the target message is a concept such as *USA*, *winter*, or *freedom*. The encoder alphabet Σ_E consists of images, while the decoder alphabet Σ_D consists of candidate concepts. The encoder selects several images from Σ_E under a multiple-choice or fixed-size signature, and the decoder maps this image set back to a concept in Σ_D . This instantiation emphasizes compositional visual encoding, since meaning arises from the relation among several images rather than from a single one.

Movie Emoji Trivia. Here, the target message is a movie title. The encoder alphabet Σ_E is a restricted set of emojis, and the encoder produces an emoji sequence under a word-sequence or fixed-length signature. The decoder then maps this representation back to a movie title in Σ_D . Compared to the image-based setting, this game places greater emphasis on metaphor, symbolic compression, and cultural association.

4 Adversarial Human–AI Captioning Task

The framework introduced above allows us to vary communicative difficulty by changing the task constraints. Here, we instantiate this idea with a captioning task over image pairs. The key variation is the *captioning condition*: in the standard condition, captions are written to support human interpretation; in the AI-aware condition, captions are written to remain as understandable as possible for humans while becoming less useful for AI. This condition may make the task harder overall, including for human receivers, but its intended effect is to reduce AI interpretability more strongly while

preserving human interpretability as much as possible. This allows us to test whether explicitly considering an AI receiver changes human encoding strategies, and whether this change reduces the alignment between human and AI interpretation.

4.1 Task Definition

To investigate selective interpretability in multimodal communication, we instantiate the framework with an adversarial captioning task over image pairs. The task is designed to study how human encoders change their captions once they are no longer asked merely to describe an image, but to communicate selectively to different kinds of receivers.

The dataset consists of 30 images organized into 15 visually related pairs. For each pair, a human participant selects one of the two images as the target message m and produces a caption c . This is done under two conditions:

- (1) **Standard Captioning Condition.** The participant writes a caption intended to allow another human to identify the selected image within the pair.
- (2) **AI-Aware Captioning Condition.** The participant writes a second caption for the same image, now under the additional instruction that the caption should remain understandable for humans while becoming harder for AI to interpret.

Formally, let m denote the selected image and let c denote the produced caption. A human decoder D_H and an AI decoder D_{AI} each select an image $\tilde{m}$ from the candidate pair. The intended selective success criterion is

$$D_H(c) = m \quad \text{and} \quad D_{AI}(c) \neq m.$$

This task operationalizes selective interpretability: the caption should act as a bridge for human receivers while becoming a hurdle for AI receivers. Importantly, participants are not instructed to produce arbitrary ambiguity or misleading captions. Instead, they are encouraged to rely on contextual nuance, abstraction, cultural reference, or pragmatic inference that they believe another human would still understand.

4.2 Dataset Construction

The 30 images were drawn from three categories:

- **Internet memes** [2] (10 images),
- **Stock photography**¹ (10 images),
- **Abstract art** (10 images).

The images were selected to provide potential for *hidden meaning*. That is, they should not only be describable through literal visual features, but should also allow for associations, moods, cultural references, or contextual interpretations. The task does not prescribe which of these meanings should be used. Instead, it is left to the human encoder to identify which aspects of an image may remain understandable to other humans while becoming harder for AI to use.

These categories were chosen because they provide different kinds of hidden-meaning potential and therefore encourage different encoding strategies.

¹<https://github.com/cj-mills/pexels-dataset>

Memes often depend on shared cultural knowledge, humor, and implicit narrative structure. They are therefore particularly relevant for studying whether humans use cultural or contextual references when they try to communicate against AI. In such cases, a caption may rely less on literal scene description and more on associations that another human is expected to recover.

Stock photographs typically depict literal and prototypical concepts with relatively clear visual cues. They therefore provide an important contrast: because the image content is often straightforward, standard captioning is expected to be easier, while the AI-aware condition becomes more challenging. If the image is already strongly anchored in a literal concept, producing a second caption that remains helpful for humans while becoming less useful for AI is harder.

Abstract art provides the opposite extreme. These images often lack explicit object-level semantics and are more open to interpretive, affective, or anecdotal readings. They are therefore useful for studying whether humans shift toward less literal and more idiosyncratic captions when obvious descriptive cues are unavailable.

To construct visually related pairs, we used different pairing criteria for the three categories. For abstract art, pairs were selected from works by the same artist in order to increase stylistic similarity while preserving interpretive ambiguity. For stock photography, images were paired when they belonged to the same search term or concept, thereby creating pairs with a high degree of literal semantic overlap. For memes, similarity is more difficult to define explicitly, so we paired memes primarily based on close CLIP embeddings [6], using embedding proximity as an approximate indicator of multimodal similarity.

By combining these three categories, the dataset covers different points on a spectrum from literal and prototypical to highly contextual or interpretive. This makes it possible to analyze not only whether captions become harder for AI, but also which kinds of image content encourage humans to use cultural, affective, metaphorical, or anecdotal forms of encoding.

Figure 4 shows one example pair from the meme category. Both images depict unusual situations in public or commercial spaces and invite humorous or anecdotal interpretation beyond their literal visual content. A human encoder may choose to describe the visible scene directly, but may also rely on social embarrassment, absurdity, or a shared “internet meme” style of interpretation in the AI-aware condition.

5 Evaluation

We evaluate the task along two dimensions: whether the intended image is selected correctly, and whether human and AI receivers interpret the same captions in aligned ways.

5.1 Evaluation Setup

We evaluate the task under both the standard and the AI-aware captioning conditions. The goal is not only to measure whether the intended image is selected correctly, but also to analyze whether humans and AI interpret the same captions in similar ways. This distinction is important because our task is not a pure accuracy benchmark. We are interested in whether the AI-aware condition changes the relation between human and AI interpretation.



Figure 4: Example of a visually related image pair from the meme category. The pair provides literal visual overlap while also allowing humorous, anecdotal, or socially contextual captions.

The captions were produced by two human annotators. Since the dataset is small, we combine all captions into a single evaluation set of 30 caption instances over the same 15 image pairs. For each caption, the selected target image serves as a reference label H_0 . We compare this label with two additional human ratings, H_1 and H_2 , and with three AI ratings, AI_1 , AI_2 , and AI_3 . In the current setup, all AI ratings are produced by repeated runs of the same multimodal model family, Gemma3-4B [9], chosen mainly for practical reasons: it can be self-hosted and integrated easily into the framework.

We report both **accuracy** and **agreement**. Accuracy measures whether a rater identifies the intended image. It therefore captures task success with respect to the selected target. Agreement, however, captures interpretive alignment. It tells us whether raters converge on the same reading of a caption, independently of whether that reading is correct. This is central to our question: if AI-aware captions act as a hurdle, they may not only reduce AI accuracy, but also reduce the alignment between human and AI interpretation.

We measure within-group agreement separately for human raters and AI raters using Fleiss’ κ [5]. For N items, n raters per item, and k categories, Fleiss’ κ is defined as

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e},$$

where

$$\bar{P} = \frac{1}{N} \sum_{i=1}^N p_i$$

is the observed agreement across items, and

$$\bar{P}_e = \sum_{j=1}^k p_j^2$$

is the agreement expected under chance, with p_j denoting the overall proportion of assignments to category j . Human Fleiss’ κ indicates how consistently human raters interpret the captions, while AI Fleiss’ κ indicates how consistent the AI system is across repeated runs.

To compare humans and AI directly, we collapse the human ratings and AI ratings into majority-vote decisions and measure

cross-group agreement using Cohen’s κ [5]:

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where p_o is the observed agreement and p_e the agreement expected by chance. This metric is especially important for our analysis because it measures whether the human and AI groups still converge on the same interpretation of a caption. A decrease in Cohen’s κ suggests that the two groups are not merely differing in accuracy, but are drifting toward different readings of the same multimodal cue.

Finally, we model label assignment statistically. Let

$$Y_{ij} = \begin{cases} 1 & \text{if rater } j \text{ assigns label } B \text{ to item } i, \\ 0 & \text{if rater } j \text{ assigns label } A. \end{cases}$$

We fit logistic models of the form

$$\text{logit}(\Pr(Y_{ij} = 1)) = \beta_0 + \beta_1 \text{AI}_{ij},$$

where $\text{AI}_{ij} = 1$ for AI ratings and $\text{AI}_{ij} = 0$ for human ratings. The coefficient β_1 tests whether AI and human raters differ systematically in their tendency to assign one of the two labels. To account for repeated observations per item, we use both a generalized estimating equation (GEE) model [8] clustered by item and a generalized linear mixed model (GLMM) [8] with an item-level random intercept.

Our expectation is that the AI-aware condition should make decoding harder for both humans and AI. However, if the introduced hurdle exploits hidden meaning that remains more accessible to humans, then humans should remain more accurate than AI, and agreement between human and AI majority decisions should weaken. Such a pattern would indicate not only a change in task success, but also a change in interpretive alignment. Changes in the qualitative behavior of captioning itself, such as a shift from literal description toward metaphor or affective association, are analyzed separately in the discussion.

5.2 Results

Table 1 summarizes the main quantitative results for the standard and AI-aware captioning conditions. Overall, the AI-aware condition made the task harder for both groups. Human accuracy dropped from 98.3% to 85.0%, and mean AI accuracy dropped from 52.2% to 43.3%. Thus, the additional hurdle affected not only AI systems, but human decoders as well.

At the same time, the results show that the two groups did not change in the same way. Human inter-rater agreement decreased substantially, with Fleiss’ κ dropping from 0.954 to 0.636. AI inter-rater agreement, in contrast, increased slightly from 0.591 to 0.679. This means that the AI-aware condition did not simply make AI behavior more random. Instead, the AI raters became somewhat more consistent with one another, even while their accuracy decreased.

Most importantly for our alignment question, agreement between human-majority and AI-majority decisions became weaker, with Cohen’s κ falling from 0.167 to -0.024 . This suggests that the AI-aware condition increased the divergence between human and AI interpretation. The same captions no longer guided both groups toward the same choices to the same extent.

Table 1: Comparison of standard and AI-aware captioning conditions. Human accuracy is averaged over H_1 and H_2 , and AI accuracy is averaged over AI_1 , AI_2 , and AI_3 .

Metric	Standard	AI-aware
Human Fleiss’ κ	0.954	0.636
AI Fleiss’ κ	0.591	0.679
Human–AI Cohen’s κ	0.167	-0.024
Mean human accuracy	98.3%	85.0%
Mean AI accuracy	52.2%	43.3%
AI majority accuracy	53.3%	40.0%
GEE odds ratio (AI vs. Human)	4.17	4.79
GEE p-value	0.0013	0.0008
GLMM odds ratio (AI vs. Human)	7.69	11.14

The logistic analyses support this pattern. In both conditions, AI ratings differed significantly from human ratings in their labeling behavior, and this difference became stronger in the AI-aware condition. In the GEE model, the estimated AI-versus-human odds ratio increased from 4.17 to 4.79, and in the GLMM from 7.69 to 11.14. These results indicate that the harder captions not only lowered AI accuracy, but also shifted AI behavior further away from human judgments.

6 Discussion: Bridge–Hurdle Effects in Adversarial Captioning

6.1 Main Findings

Table 1 summarizes the main results for the standard and AI-aware captioning conditions. Overall, the adversarial captioning task supports the bridge–hurdle idea only partially, but it reveals a more nuanced effect: introducing the hurdle does not simply reduce performance uniformly. Instead, it changes the relation between human and AI interpretation.

Under the AI-aware captioning condition, decoding performance decreased for both groups. Mean human accuracy dropped from 98.3% to 85.0%, while mean AI accuracy dropped from 52.2% to 43.3% (Table 1). This shows that the introduced hurdle is a real communicative hurdle for all receivers, not only for AI. At the same time, however, the results do not suggest a simple collapse of AI decoding. Rather, the harder captions appear to change *how* AI fails.

Most notably, agreement between the human-majority and AI-majority decisions became worse in the AI-aware condition, with Cohen’s κ dropping from 0.167 to -0.024 (Table 1). At the same time, agreement among the AI raters themselves increased slightly, with Fleiss’ κ rising from 0.591 to 0.679, whereas human inter-rater agreement decreased substantially from 0.954 to 0.636. In other words, AI raters became more consistent with one another, but less aligned with human judgments.

This pattern suggests that the AI-aware captions do not merely make AI behavior noisier or more random. Instead, they appear to push AI systems toward a more stable but less human-like interpretation strategy. The more important effect of the hurdle is not

lower AI performance alone, but the increased divergence between human and AI decoding.

This interpretation is also consistent with the logistic analyses reported in Table 1. In both conditions, AI ratings differed significantly from human ratings in their tendency to assign one of the two labels, and this difference became stronger in the harder condition. In the GEE model, the estimated odds ratio increased from 4.17 to 4.79, and in the GLMM from 7.69 to 11.14. These results therefore do not directly measure correctness, but they indicate a systematic difference in labeling behavior. Together with the accuracy and agreement results, they suggest that the AI-aware captions change the success rates and the decision tendencies.

At the same time, the results do *not* fully support the strongest possible version of our initial hypothesis. We expected that introducing the hurdle would weaken agreement more strongly for AI than for humans. Instead, human inter-rater agreement decreased much more strongly than AI inter-rater agreement (Table 1). This means that the harder captions also reduced the invariance of human interpretation. Put differently, the captions did not act as a pure bridge for humans and a pure hurdle for AI; they functioned as a hurdle for both groups, but in different ways. For humans, the hurdle appears to increase ambiguity and reduce shared interpretation. For AI, it appears to increase separation from human interpretation without necessarily increasing internal inconsistency.

6.2 Qualitative Observations and Interpretation

The quantitative results become easier to interpret when viewed together with the observed changes in captioning style. These interpretations are qualitative and should be read as plausible explanations of the observed patterns rather than as independently verified causal mechanisms. In the standard condition, captions tend to be longer, more literal, and more descriptive of visible scene content. In the AI-aware condition, they often become shorter and more indirect, relying more heavily on mood, connotation, metaphor, or anecdotal association. Figure 5 illustrates this shift: the standard caption describes visible scene elements: *Forest and grass with sunshine, some shadow in the foreground and in the background more dense forest*, whereas the AI-aware caption reduces the image to the evocative label: *Fairy Tale*. This may indicate that human participants do not merely try to obscure the image; rather, they may shift from describing *what is visible* to describing *what the image feels like* or *what it evokes*.

This observation is important because it reveals not only possible weaknesses of the AI system, but also the human sender’s expectations about those weaknesses. In the AI-aware condition, participants must form an implicit model of what the AI is likely to understand or misunderstand. The resulting captions therefore provide evidence about a human theory of AI limitations. Qualitatively, the captions suggest that participants expect AI systems to perform relatively well on literal object- or scene-level description, but less well on affective, metaphorical, or pragmatically grounded meanings.

A second qualitative observation concerns the role of AI priors. In some cases, the AI appears to prefer images or concepts that are highly salient in its pretraining data, even when the caption points elsewhere. For example, in the pair shown in Figure 6, the AI

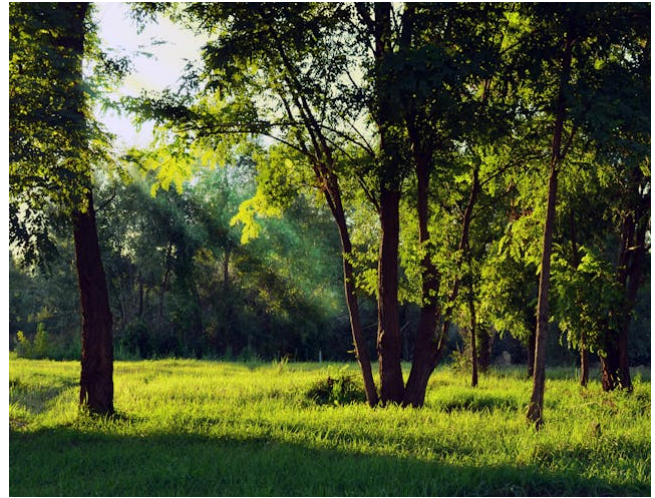


Figure 5: Image at which we see a clear caption shift from literal description to a shorter, more evocative label in the AI-aware condition.

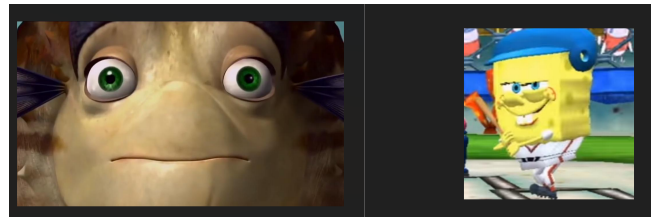


Figure 6: Example pair where a highly recognizable concept (here SpongeBob) may bias AI interpretation.

tended to favor the image containing SpongeBob, even when the caption was intended to guide interpretation toward the alternative image. This suggests that strong prior familiarity may sometimes dominate the decoding process.

From the perspective of multimodal communication, this is highly relevant: the AI does not only interpret the given caption, but also brings strong learned expectations about which visual concepts are most likely or most recognizable. Such priors may act as a hidden bias in the opposite direction, making some images more attractive to the AI system because they are strongly represented in its training distribution or internal representations. The hurdle therefore does not operate in a neutral interpretive space. Instead, it interacts with asymmetric background knowledge: for humans, this may take the form of shared culture, pragmatic context, or everyday experience; for AI systems, it may take the form of pretraining frequency, stereotypical visual associations, or strong embedding priors.

Taken together, these qualitative observations refine the interpretation of the quantitative results. The AI-aware captions do not simply make the task harder in a uniform way. Rather, they shift the kinds of cues that are foregrounded in communication. Humans appear to rely more on indirect, affective, or anecdotal cues, while AI systems may remain more dependent on literal scene content

or on highly salient pretrained concepts. This helps explain why the harder condition reduced performance for both groups, yet at the same time increased the divergence between human and AI interpretation.

6.3 Limitations

A few limitations should be kept in mind when interpreting the results. First, the dataset is small. With only 30 caption instances over 15 image pairs, the quantitative findings in Table 1 should be treated as exploratory rather than definitive. Small samples make agreement estimates less stable and limit how far the observed effects can be generalized.

In addition, the qualitative interpretations of captioning strategies and AI priors are based on observed examples rather than on a separate annotation scheme. They should therefore be treated as exploratory explanations that motivate further analysis, not as conclusive evidence about the underlying causes of individual model decisions.

Second, the human reference label H_0 is both the target choice and part of the human-side rating setup. This is useful for organizing the evaluation, but it also means that the human and AI sides are not fully symmetric.

Third, the three AI raters are repeated runs of the same multimodal model family rather than different model families. Their agreement should therefore be interpreted primarily as a form of model self-consistency, not as agreement across heterogeneous AI systems.

Fourth, agreement coefficients such as Fleiss’ κ and Cohen’s κ are sensitive to label prevalence. Changes in category balance may therefore influence the absolute values of these measures in addition to genuine changes in interpretive behavior.

Fifth, the logistic model of label assignment should be interpreted cautiously. The model tests whether AI and human raters differ in their tendency to assign label B rather than label A , but it does not directly model correctness. Therefore, the reported odds ratios capture systematic differences in labeling behavior or possible label bias, not whether the AI-aware condition selectively reduced AI accuracy. This analysis is useful for identifying shifts in decision tendencies, but it should be treated as complementary to the accuracy and agreement measures.

Finally, the present study focuses on one specific multimodal setting, namely paired-image captioning. While this is sufficient to demonstrate the framework and to probe bridge–hurdle effects, it does not yet show how far the same patterns extend to other multimodal games, modalities, or model families.

6.4 Implications and Future Work

Taken together, the results suggest that adversarial captioning is a useful way to study multimodal interpretation beyond simple benchmark-style accuracy. The main effect of the AI-aware condition is not only that AI performance decreases, but that human and AI interpretations become less aligned. In this sense, the task reveals not just whether communication succeeds, but how differently the same multimodal cue may be understood by different receivers.

A second contribution of this work is methodological. We introduced a general framework for representing multimodal games through a shared sender–receiver–validator structure. This framework is not limited to the captioning task studied here, but is intended to capture a broader class of multimodal communication games. By separating tasks, alphabets, signatures, adapters, and agents, it provides a reusable way to compare different game settings within one common execution logic.

Within this framework, the bridge–hurdle distinction can be studied in a controlled way through *difficulty*. Rather than treating communicative difficulty as an accidental by-product, the framework turns it into a design parameter. By varying the available alphabets, the admissible signatures, or the participating agents, one can steer whether multimodality primarily supports interpretation or obstructs it. In this sense, the framework helps isolate the bridge–hurdle aspect of multimodal communication more explicitly than task-specific analyses alone.

The qualitative observations refine this picture further. Human participants appear to shift from literal scene description toward shorter, more evocative, or more associative captions when they try to communicate against AI. The resulting captions therefore provide a window onto not only possible AI limitations, but also human expectations about those limitations. At the same time, the SpongeBob example suggests that AI interpretation is shaped not only by the caption itself, but also by strong priors tied to highly familiar visual concepts. Bridge–hurdle effects therefore emerge from the interaction between message, modality, task design, and receiver-specific background knowledge.

More broadly, this supports the central claim of the paper: multimodality does not have a fixed role as either bridge or hurdle. Rather, its role depends on who is interpreting the message and which cues are available or salient to that receiver.

Future work should therefore proceed in three directions. First, the dataset should be expanded to include more image pairs, more annotators, and more diverse participants. Second, the AI side should be broadened to include multiple model families rather than repeated runs of a single one. Third, the framework itself should be reused across additional multimodal games beyond paired-image captioning, so that bridge–hurdle effects can be compared systematically across different modalities, symbol sets, and task mechanics. Such extensions would make it possible to study more precisely which kinds of hidden meaning remain accessible to humans while becoming a hurdle for AI.

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References

- [1] Paul Copley and Litza Jansz. 1998. *Semiotics for beginners*. Icon Books (Penguin Group).
- [2] Niklas Deckers, Aleya Albayrak, Klara M. Gutekunst, Paul Hahn, Henry Plutz, Moritz Wirth, and Martin Potthast. 2026. A Resource for Evaluating Creative Search on the Infinite Index. (2026). Unpublished manuscript.
- [3] Umberto Eco. 2002. *Einführung in die Semiotik*. utb GmbH.
- [4] Sean Hall. 2007. *This Means This, This Means That: A User’s Guide to Semiotics*.
- [5] Klaus Krippendorff. 2018. *Content analysis: An introduction to its methodology*. Sage publications.
- [6] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al.

2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PmlR, 8748–8763.

[7] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. 2021. Zero-shot text-to-image generation. In *International conference on machine learning*. Pmlr, 8821–8831.

[8] Walter W Stroup, Marina Ptukhina, and Julie Garai. 2024. *Generalized linear mixed models: modern concepts, methods and applications*. Chapman and Hall/CRC.

[9] Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivière, Mihir Sanjay Kale, Juliette Love, et al. 2024. Gemma: Open models based on gemini research and technology.

arXiv preprint arXiv:2403.08295 (2024).

A Use of AI Tools

AI tools such as DeepL and Perplexity were used for language support, proofreading, translation, and background research. All core ideas, methodological decisions, and conclusions of this work are the authors’ own.